

AI Shopping Agent: Capabilities and Knowledge Base Architecture

This document outlines the specific operational tasks the AI E-Commerce Shopping Agent can execute, alongside the underlying knowledge base and data structures it utilizes to make informed, highly relevant product recommendations.

1. Agent Capabilities (Task Execution)

The AI agent acts as a full-stack personal shopper. Its capabilities are divided into four main phases: Interaction, Extraction, Evaluation, and Presentation.

1.1 User Interaction & Intent Extraction

- **Natural Language Processing (NLP):** Parses complex, multi-layered user queries (e.g., "I need a vegan leather jacket for winter, under \$150, that arrives by Friday").
- **Clarification Prompting:** If a user's request is too broad (e.g., "Find me a laptop"), the agent will ask targeted follow-up questions to narrow down requirements (gaming vs. work, screen size, budget).
- **Preference Profiling:** Saves user constraints (like sizing, preferred brands, or geographical location) for future sessions.

1.2 Web Traversal & Data Ingestion

- **Multi-Channel Searching:** Executes concurrent searches across target marketplaces (Amazon, Walmart, BestBuy, specialty stores).
- **Dynamic Filtering:** Applies initial hard filters during the search phase to eliminate irrelevant items immediately (e.g., filtering out out-of-stock items or items above the strict budget limit).
- **Review & Q&A Scraping:** Fetches the top helpful reviews, critical reviews, and customer Q&A sections for the candidate products.

1.3 Evaluation & Decision Making

- **Sentiment Analysis:** Reads customer reviews to determine the real-world consensus on durability, sizing accuracy, and build quality.
- **Feature Normalization:** Standardizes specifications across different sites (e.g., converting cm to inches, or equating "graphite" to "dark grey") so products can be compared apples-to-apples.
- **Fake Review Detection:** Flags products with highly suspicious review patterns (e.g., hundreds of 5-star reviews submitted on the same day).
- **Scoring & Ranking:** Runs the aggregated data through a customized scoring matrix to rank the top 5 options.

1.4 Post-Decision Capabilities

- **Price Drop Alerts (Optional):** If the user decides not to buy, the agent can monitor the top 5 items and alert the user if the price drops.
- **Alternative Suggestions:** If no items perfectly match the criteria, the agent can suggest compromises (e.g., "I couldn't find a 4K monitor under \$100, but here are the best 1080p options, or the cheapest 4K option at \$150").

2. The Knowledge Base (The "Brain")

Unlike a traditional chatbot, this agent relies on a hybrid knowledge base: a mix of static pre-trained knowledge, real-time dynamic data, and user-specific memory.

2.1 Real-Time Contextual Data (Dynamic)

This is the most critical part of the knowledge base. The agent cannot rely on its training data for shopping, as prices and stock change daily.

- **Live Product Catalogs:** Real-time JSON feeds containing current price, stock status, seller ratings, shipping times, and variants (colors/sizes).
- **Live Review Datasets:** Fresh text data pulled directly from the product pages to assess current batch quality.

2.2 Pre-Trained Domain Knowledge (Static)

The Large Language Model (LLM) brings inherent understanding of the world, which it uses to interpret features.

- **Material & Quality Ontology:** The agent "knows" that *merino wool* is generally higher quality and warmer than *acrylic*, or that *OLED* screens have deeper blacks than *IPS* screens.
- **Brand Positioning:** Understanding that "Herman Miller" implies premium/expensive, while "Amazon Basics" implies budget-friendly/functional.
- **Semantic Synonyms:** Knowing that a user asking for a "sturdy" table is looking for keywords like "solid wood," "steel frame," or "high weight capacity" in the product specs.

2.3 User Profile & Memory (Personalized)

- **Explicit Constraints:** Hard rules set by the user (e.g., "Never show me products from Brand X," or "Max budget is strictly \$500").
- **Implicit Preferences:** Learned behaviors over time (e.g., The user consistently chooses the cheapest option among the top 5, so the agent might start weighing 'Value for Money' slightly higher for this specific user).

2.4 Rules Engine & Scoring Matrices (Logic)

- **The Evaluation Matrix:** The weighted formula (Detailed in the PRD) used to calculate the

final score of a product.

- **Red Flag Rules:** Hardcoded logic to immediately disqualify products (e.g., *IF* average rating < 3.0, *THEN* discard; *IF* seller rating < 80%, *THEN* discard).

3. Task to Knowledge Base Mapping

Agent Task	Primary Knowledge Base Used	Output/Result
Parsing User Prompt	Pre-Trained Domain Knowledge (NLP, Synonyms)	Structured JSON of search parameters
Filtering Bad Products	Rules Engine (Red Flag Rules) + Live Catalogs	Shortlist of viable candidates (Top 20)
Assessing Quality	Live Review Datasets + Pre-Trained Ontology	Sentiment Score & Pros/Cons list
Final Selection	Evaluation Matrix + User Profile constraints	The Final Top 5 List